NUANCE: Natural-language Understanding of Ambiguous Navigation Commands in Environments

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Abstract

Human-in-the-loop robotics is important for critical systems such as autonomous drones or search-and-rescue. However, ambiguous instructions make this difficult as there are no fixed numeric meaning and a robot must learn them from the context. We propose learning this mapping from human raters analyzing robot motion. A Bostom Dynamics Spot reaches a decision point in an indoor scene, receives a vague instruction, and executes candidate actions. Human raters score these clips from 1–5 under the original instruction and alternatives, and a reward model predicts the scores. A Qwen3.5-4B vision-language policy with LoRA adapters is optimized against the reward model with Group Relative Policy Optimization. We compare this against the base model, supervised fine-tuning, and DPO with human ratings.

1 Introduction

Autonomous robotics is highly important for critical systems, such as autonomous drones, driving, or search-and-rescue, but is dangerous in high pressure situations, where each stochastic decision compounds the error rate until the output is misaligned from the human goals (Ross et al., 2011). Human-in-the-loop (HITL) reduces this by ensuring decisions are aligned with the operator's goals, but model misunderstandings may lead to catastrophic outcomes. Thus, we propose NUANCE to improve robot understanding for decision making in HITL systems to improve downstream safety and decision making.

1.1 Research Question

Our research investigates the following question: Does NUANCE improve robot understanding and decision making under ambiguous natural language instructions? We hypothesize GRPO against the learned reward model will achieve higher human ratings than all three baselines on vague instructions for unseen scenes while matching on unambiguous instructions.

1.2 Motivation

HITL practices can be resource-intensive, and sometimes that resource cost is not acceptable in real-world scenarios. Generally, HITL adds a time overhead where humans must review actions, provide feedback, and/or clarify commands. This involves real human labor and is not scalable for most robotic applications. Minimizing the need for HITL practices by improving the robot's comprehension of human intentions can help surmount these resource constraints and allow robotic applications to be used autonomously at scale.

Alignment is another pressing issue, increasingly concerning as AI systems get smarter and more robust, and improving understanding of human intentions will help solve it. Being able to leave autonomous systems alone for long periods of time without having to worry about them taking actions that are obviously harmful to humans promotes greater trust in autonomous systems and reduced need for human oversight.

2 Related Work

"LATTE: LAnguage Trajectory TransformEr" by Bucker et al. leverages BERT and CLIP embeddings as inputs to a simple transformer-MLP model in order to translate qualitative modifiers into quantitative robot behaviors. It uses natural language commands, scene geometry, and environment images as raw inputs, outputting modified robot trajectories. However, LATTE learns modifiers from procedurally generated trajectories so the vague language has predetermined numeric meanings while it is context-dependent in NUANCE. (Bucker et al., 2022)

"Grounding Robot Plans from Natural Lan-

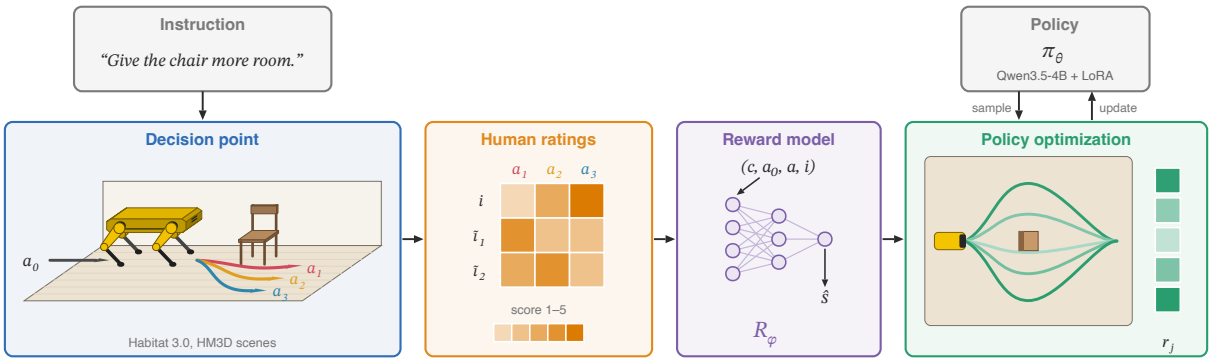


Figure 1: Experimental pipeline from human-rated candidate actions to GRPO training against the learned reward model, with illustrative rating and reward values.

guage Instructions with Incomplete World Knowledge" by Nyga et al. uses probabilistic relational reasoning to infer symbolic robot plans from natural language instructions, perceived world states, and background knowledge, accounting for missing information about objects and task requirements. (Nyga et al., 2018)

"LocoVLM: Grounding Vision and Language for Adapting Versatile Legged Locomotion Policies" by Nahrendra et al. lets a legged robot adjust how it walks in real time from human feedback. An LLM builds a dataset of locomotion skills and instructions, and a VLM then builds a semantic representation which is matched against the database to recommend an action. However, it chooses from a fixed set of skills rather than resolving vague context-dependent language. (Nahrendra et al., 2026)

"SayTap: Language to Quadrupedal Locomotion" by Tang et al. uses the foot contact patterns of a quadruped as the interface between a human's natural language commands and the controller. An LLM writes a sequence of 1s and 0s and marks when each foot hits the ground or is in the air. An RL controller then executes on the robot. However, speed commands are mapped onto a fixed set of target velocities so meaning does not change based on context. (Tang et al., 2023)

Unlike prior work, NUANCE's novelty comes from being the first approach to learn context-dependent meaning of vague instructions from human ratings of robot motion.

3 Proposed Approach

We consider each clip with 3 decision points (Figure 1), where the model is given a setting such as a room c, an action a_0 it was performing, and an instruction i to execute. The model then chooses an action a, such as where to go, how fast to walk, or how much room to leave around obstacles as a structured command executed by a low-level controller.

After collecting N clips in a simulator, we employ human raters to score each clip on a 5-point scale for how well it matches the given instruction. We rate each clip under different instructions, so the instruction "go a little faster" might be paired with the clip "go much faster" to compare the same motion against different instructions for determining the commanded difference from the semantic meaning (how much faster is "a little faster" compared to "much faster").

The robot is a Boston Dynamics Spot robot with an on-board Jetson Orin NX lent from a Purdue lab used for research. Compute for training will use NVIDIA A10 nodes on Purdue's Gilbreth cluster with evaluations and simulations running on an M5 Mac Mini.

The policy is a Vision Language Model (VLM) to map the robot's sensors to the next action. At each decision point, it sees the front image, instruction, and a grounded text description of the current motion. The VLM then outputs an action with a goal point and speed. The goal point is then projected onto the 3D map using Spot's WorldObjectService, and the low-level controller executes the action. We use a VLM rather than pretraining a policy as the same instruction has different meanings depending on the context, and a pretrained VLM handles the room context understanding. Additionally, a pretrained VLM understands spatial phrases without task-specific training.

We train the policy with Group Relative Policy Optimization (GRPO) (Shao et al., 2024), with

lightweight LoRA adapters (Hu et al., 2022) on the VLM. For each context (room c, instruction i pairs) the policy proposes several actions, which are then scored, and the policy is updated towards successful actions while ignoring unsuccessful actions. As a result, a simulator is only used for generating clips for training and evaluation rather than in the training loop.

4 Experimental Plan

We use the Habitat 3.0 simulator (Puig et al., 2024) for HITL capabilities with a simulated Spot robot on indoor scenes from the Habitat-Matterport 3D dataset (HM3D) (Ramakrishnan et al., 2021). We choose 12 scenes consisting of corridors, open rooms, cluttered rooms, and rooms with simulated humans walking through them. The robot is given a predetermined 2D path and all pathfinding is handled by Spot.

In each scene, a decision point is made along the line with a vague instruction. Instructions are vague spatial goals such as "go to the empty space", modifiers such as "go a little faster", or clearance requests like "give the chair more room". We additionally include unambiguous instructions such as "go to the purple chair" as a control as our advantage should be on vague language. For each decision point, we render three candidate actions across plausible goals and speeds. We estimate rendering a dataset of 2000 clips, with a standard 70/10/20 train, validation, test split.

The two researchers rate the clips and score each from 1–5 under the original instructions and two alternatives, with a subset of 200 clips rated by both to determine inter-rater variability. We estimate a total of five hours is required for rating clips.

Once the clips are rated, we train a small reward model to predict the human scores for any (c, a_0, a, i) tuple. We check against held-out ratings before using the policy. The policy is Qwen3.5-4B (Qwen Team, 2026) with LoRA adapters. We begin the recipe with supervised warm-up on highly-rated actions to teach the output format. Then, GRPO (implemented with HuggingFace's TRL Library (von Werra et al., 2020)) optimizes against the reward model.

We compare our policy against the following alternatives on the same model and data:

1. The base VLM with no post-training
2. Supervised fine-tuning on highly-rated actions
3. DPO (Rafailov et al., 2023) on preference pairs from the ratings

We evaluate with two data sources. First, we use unseen scenes of room types that were seen in training. We then compare with unseen scenes of new room types to test whether the policy generalizes to new spaces. Human raters then score these clips, where we measure the average human ratings by instruction type and split. Then, we run the best policy on the Spot aboard the Jetson Orin NX in all four scenes at the Hicks Underground Library Lab Space as a qualitative final test.

5 Conclusion

Ambiguous natural language is a major obstacle for HITL robotics, as improved understanding allows for reducing required human oversight and better decision making. This proposal addresses this problem by post-training a VLM against a reward model from learned human ratings, and evaluates the policy against other methods in both simulation and real life performance to make HITL systems both more scalable and safer to deploy.

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